



Bahria University, Islamabad

Department of Software Engineering

Computer Communication & Networking Lab
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Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1					
2					
3					
4					
5					

Comments:

Signature

LAB # 9

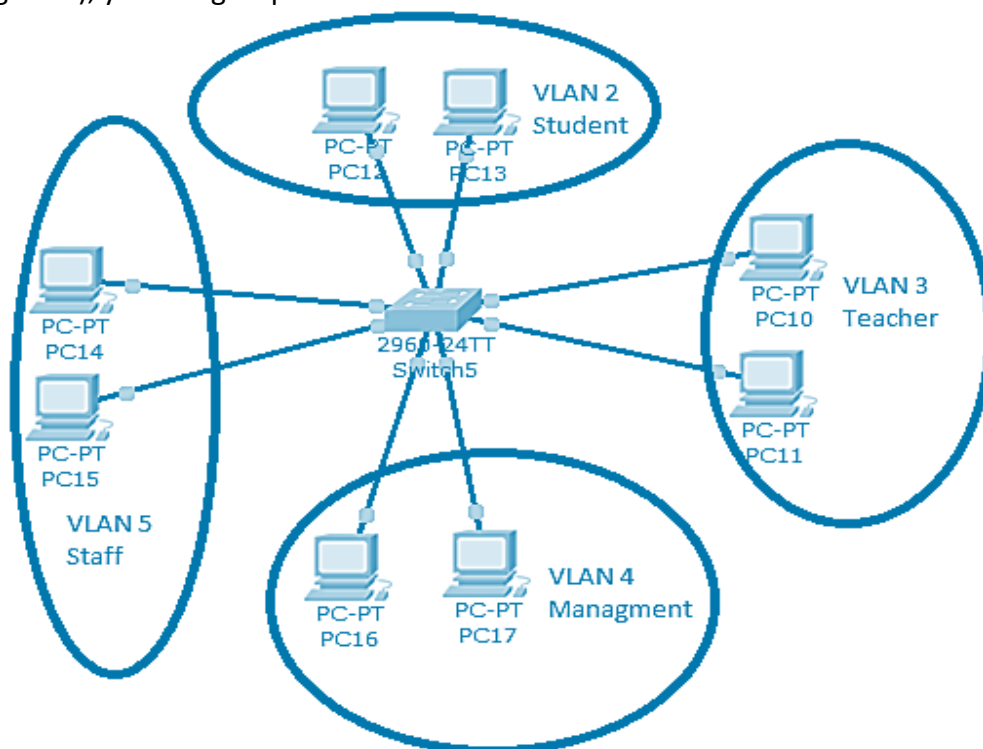
Building and Testing VLAN Network

Objective

To familiarize students with the design and construction of VLAN, and also with the tools that can be used to check the network

Introduction

A VLAN is a switched network that is logically segmented by functions, project teams, or applications without regard to the physical location of the users. For example several end stations might be grouped as a department, such as engineering or accounting. When the end stations are physically located close to one another, you can group them into a LAN segment. If any of the end stations are in different buildings (not the same physical LAN segment), you can group them into a VLAN.



VLAN Ranges (Cisco Standard)

- VLAN 0 & 4095: Reserved, not usable.
- VLAN 1: Default VLAN; all switch ports belong here initially. Cannot be deleted.
- VLAN 2–1001: Normal VLAN range (configurable, editable, deletable).
- VLAN 1002–1005: Reserved for legacy Token Ring and FDDI.
- VLAN 1006–4094: Extended VLAN range.

Problem

Designing VLAN

Equipment

1. Switch,
2. Necessary cables
3. PC

Procedure:

Step 1: Create VLANs on switch S1.

Use the **vlan** *vlan-id* command in global configuration mode to add a VLAN to switch S1. There are four VLANs configured for this lab: VLAN 2 (Student); VLAN 3 (Teacher); VLAN 4 (Management); and VLAN 5 (Staff). After you create the VLAN, you will be in vlan configuration mode, where you can assign a name to the VLAN with the **name** *vlan name* command.

```
S1(config)#vlan 2
S1(config-vlan)#name Student
S1(config-vlan)#vlan 3
S1(config-vlan)#name Teacher
S1(config-vlan)#vlan 4
S1(config-vlan)#name management
S1(config-vlan)#vlan 5
S1(config-vlan)#name staff
S1(config-vlan)#end
S1#
```

Step 2: Verify that the VLANs have been created on S1.

Use the **show vlan brief** command to verify that the VLANs have been created.

```
S1#show vlan brief
```

```
VLAN Name Status Ports
```

```
-----
1 default active Fa0/1, Fa0/2, Fa0/4, Fa0/5
Fa0/6, Fa0/7, Fa0/8, Fa0/9
Fa0/10, Fa0/11, Fa0/12, Fa0/13
Fa0/14, Fa0/15, Fa0/16, Fa0/17
Fa0/18, Fa0/19, Fa0/20, Fa0/21
Fa0/22, Fa0/23, Fa0/24, Gi0/1
Gi0/2
2 student active
3 teacher active
4 management active
5 staff active
```

Step 3: Assign switch ports to VLANs

Refer to the port assignment table on page 1. Ports are assigned to VLANs in interface configuration mode, using the **switchport access vlan** *vlan-id* command. You can assign each port individually or you can use the **interface range** command to simplify this task, as shown here. Save your configuration when done.

```
S1(config)#interface range fa0/1-4
```

```

S1(config-if-range)#switchport access vlan 2
S1(config-if-range)#interface range fa0/5-8
S1(config-if-range)#switchport access vlan 3
S1(config-if-range)#interface range fa0/9-12
S1(config-if-range)#switchport access vlan 4
S1(config-if-range)#interface range fa0/13-16
S1(config-if-range)#switchport access vlan 5

S1(config-if-range)#end
S1#copy running-config startup-config
Destination filename [startup-config]? [enter]
Building configuration...
[OK]

```

Step 4: Assign IP to all PC

Assign IP address to PC with same network address in same VLAN

Step 5: verify connection

Use command prompt to verify connection. Apply ping command to computer in same VLAN and in different VLAN. Which one is not successful?

In this network, each VLAN acts like a separate and isolated network. So, computers that belong to the **same VLAN** should be able to communicate with each other without any problem. This means that if I ping a PC that is in the same VLAN and same subnet, the ping **should be successful**, because the switch forwards traffic within the same broadcast domain.

However, when I try to ping a computer that is in a **different VLAN**, the ping will **not succeed**. This is because different VLANs are completely isolated from each other at Layer 2, and a normal switch cannot route traffic between them. Inter-VLAN communication only becomes possible when a router or a Layer 3 switch is added to perform routing between VLANs.

So in summary:

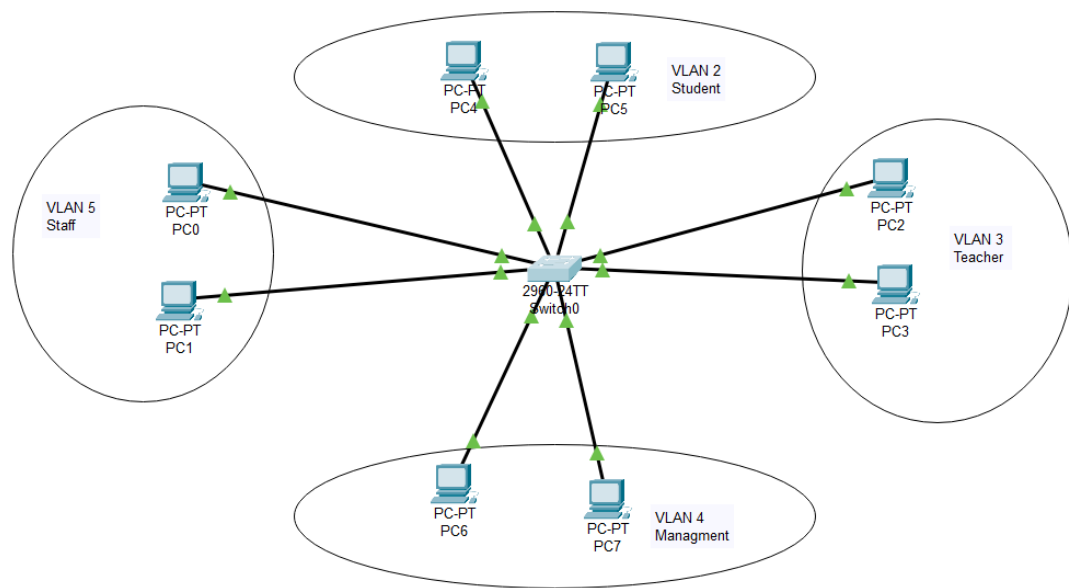
- **Ping within the same VLAN → works**
- **Ping between different VLANs → does not work**

This is the expected and correct behavior for a Layer 2 switch setup without inter-VLAN routing

Task:

1. Perform all the steps explained above and answer the questions that are being asked. Provide screen shots of results or observations where applicable.

STEP 1:



```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#vlan 2
S1(config-vlan)#name Student
S1(config-vlan)#exit
S1(config)#vlan 3
S1(config-vlan)#name Teacher
S1(config-vlan)#exit
S1(config)#vlan 4
S1(config-vlan)#name management
S1(config-vlan)#exit
S1(config)#vlan 5
S1(config-vlan)#name staff
S1(config-vlan)#exit
S1(config)#exit
S1#
%SYS-5-CONFIG_I: Configured from console by console
```

STEP 2:

```
S1#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
2 Student	active	
3 Teacher	active	
4 management	active	
5 staff	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
S1#
```

STEP 3:

```
S1#
S1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#interface range fa0/1-4
S1(config-if-range)#switchport access vlan 2
S1(config-if-range)#exit
S1(config)#interface range fa0/5-8
S1(config-if-range)#switchport access vlan 3
S1(config-if-range)#exit
S1(config)#interface range fa0/9-12
S1(config-if-range)#switchport access vlan 4
S1(config-if-range)#exit
S1(config)#interface range fa0/13-16
S1(config-if-range)#switchport access vlan 5
S1(config-if-range)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S1#
```

STEP 4:

```
Device Name: PC4
Device Model: PC-PT
```

Port	Link	IP Address
FastEthernet0	Up	10.2.0.10/8
Bluetooth	Down	<not set>

```
Device Name: PC5
Device Model: PC-PT
```

Port	Link	IP Address
FastEthernet0	Up	10.2.0.11/8
Bluetooth	Down	<not set>

```
Device Name: PC2
Device Model: PC-PT
```

Port	Link	IP Address
FastEthernet0	Up	10.3.0.10/8
Bluetooth	Down	<not set>

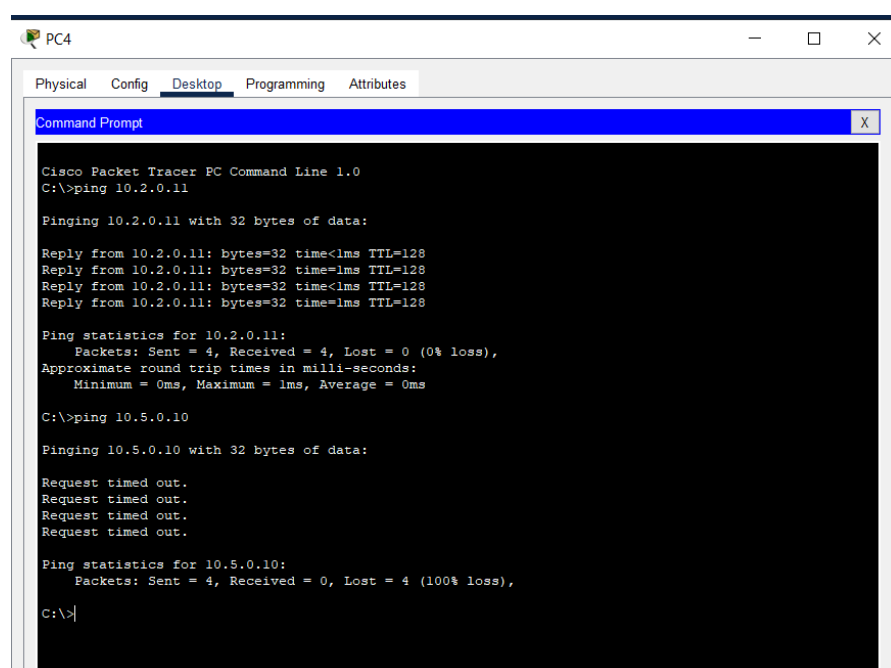
```
Device Name: PC3
Device Model: PC-PT
```

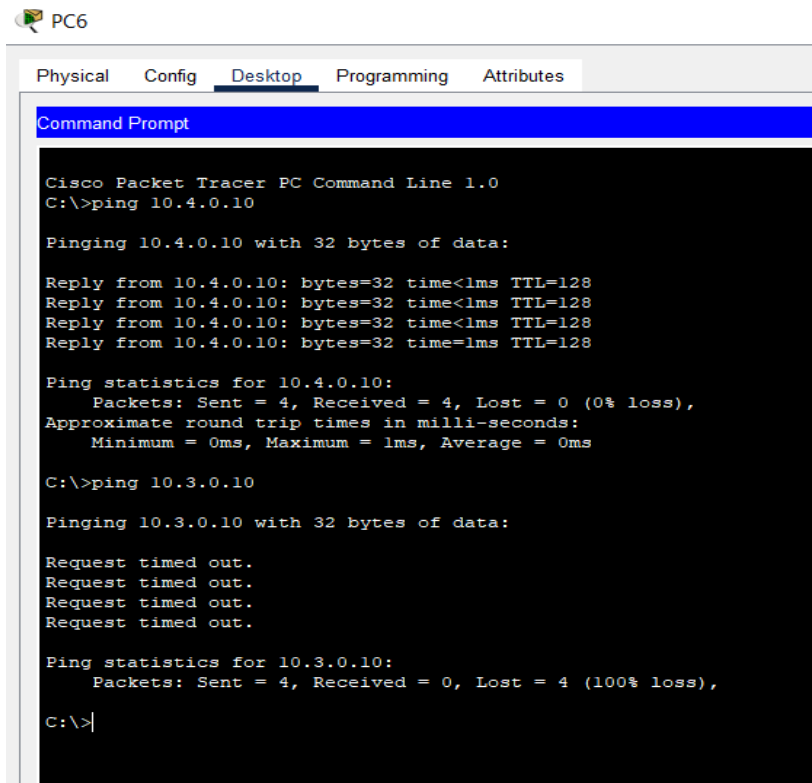
Port	Link	IP Address
FastEthernet0	Up	10.3.0.11/8
Bluetooth	Down	<not set>

Device Name: PC7			Device Name: PC6		
Device Model: PC-PT			Device Model: PC-PT		
Port	Link	IP Address	Port	Link	IP Address
FastEthernet0	Up	10.4.0.10/8	FastEthernet0	Up	10.4.0.11/8
Bluetooth	Down	<not set>	Bluetooth	Down	<not set>

Device Name: PC0			Device Name: PC1		
Device Model: PC-PT			Device Model: PC-PT		
Port	Link	IP Address	Port	Link	IP Address
FastEthernet0	Up	10.5.0.10/8	FastEthernet0	Up	10.5.0.11/8
Bluetooth	Down	<not set>	Bluetooth	Down	<not set>

STEP 5:





PC6

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.4.0.10

Pinging 10.4.0.10 with 32 bytes of data:

Reply from 10.4.0.10: bytes=32 time<1ms TTL=128
Reply from 10.4.0.10: bytes=32 time<1ms TTL=128
Reply from 10.4.0.10: bytes=32 time<1ms TTL=128
Reply from 10.4.0.10: bytes=32 time=1ms TTL=128

Ping statistics for 10.4.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.3.0.10

Pinging 10.3.0.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

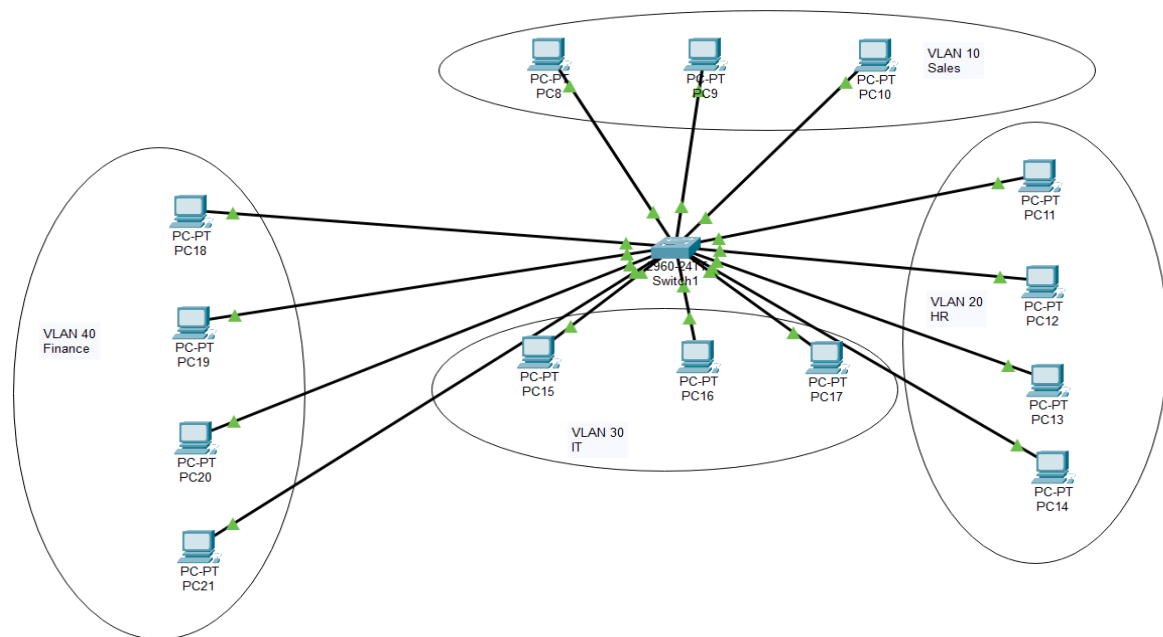
Ping statistics for 10.3.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

2. Using the concepts learned in this lab, create four VLANs on switch S1 named Sales, HR, IT, and Finance. Assign appropriate switch ports to each VLAN as per your own logical design (minimum 3 ports per VLAN). Configure IP addresses on the PCs within each VLAN and verify communication within the same VLAN and between different VLANs.

Provide screenshots of:

- a. VLAN creation (show vlan brief)
- b. Port assignments
- c. Successful and unsuccessful ping results



```

S1#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Sales	active	
20	HR	active	
30	IT	active	
40	Finance	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

S1#

```

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```

S1>enable
S1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#vlan 10
S1(config-vlan)#name Sales
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#name HR
S1(config-vlan)#exit
S1(config)#vlan 30
S1(config-vlan)#name IT
S1(config-vlan)#exit
S1(config)#vlan 40
S1(config-vlan)#name Finance
S1(config-vlan)#exit
S1(config)#exit
S1#
%SYS-5-CONFIG_I: Configured from console by console

```

```

S1#
S1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#interface range fa0/1-3
S1(config-if-range)#switchport access vlan 10
S1(config-if-range)#exit
S1(config)#interface range fa0/4-7
S1(config-if-range)#switchport access vlan 20
S1(config-if-range)#exit
S1(config)#interface range fa0/8-10
S1(config-if-range)#switchport access vlan 30
S1(config-if-range)#exit
S1(config)#interface range fa0/11-14
S1(config-if-range)#switchport access vlan 40
S1(config-if-range)#exit
S1(config)#

```

```

S1#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	Sales	active	Fa0/1, Fa0/2, Fa0/3
20	HR	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7
30	IT	active	Fa0/8, Fa0/9, Fa0/10
40	Finance	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

S1#

```

Copy

Paste

VLAN 10 – Sales

- PC8 → 192.168.10.10
 - PC9 → 192.168.10.11
 - PC10 → 192.168.10.12
- Mask: 255.255.255.0

VLAN 20 – HR

- PC11 → 192.168.20.10
 - PC12 → 192.168.20.11
 - PC13 → 192.168.20.12
 - PC14 → 192.168.20.13
- Mask: 255.255.255.0

VLAN 30 – IT

- PC15 → 192.168.30.10
 - PC16 → 192.168.30.11
 - PC17 → 192.168.30.12
- Mask: 255.255.255.0

VLAN 40 – Finance

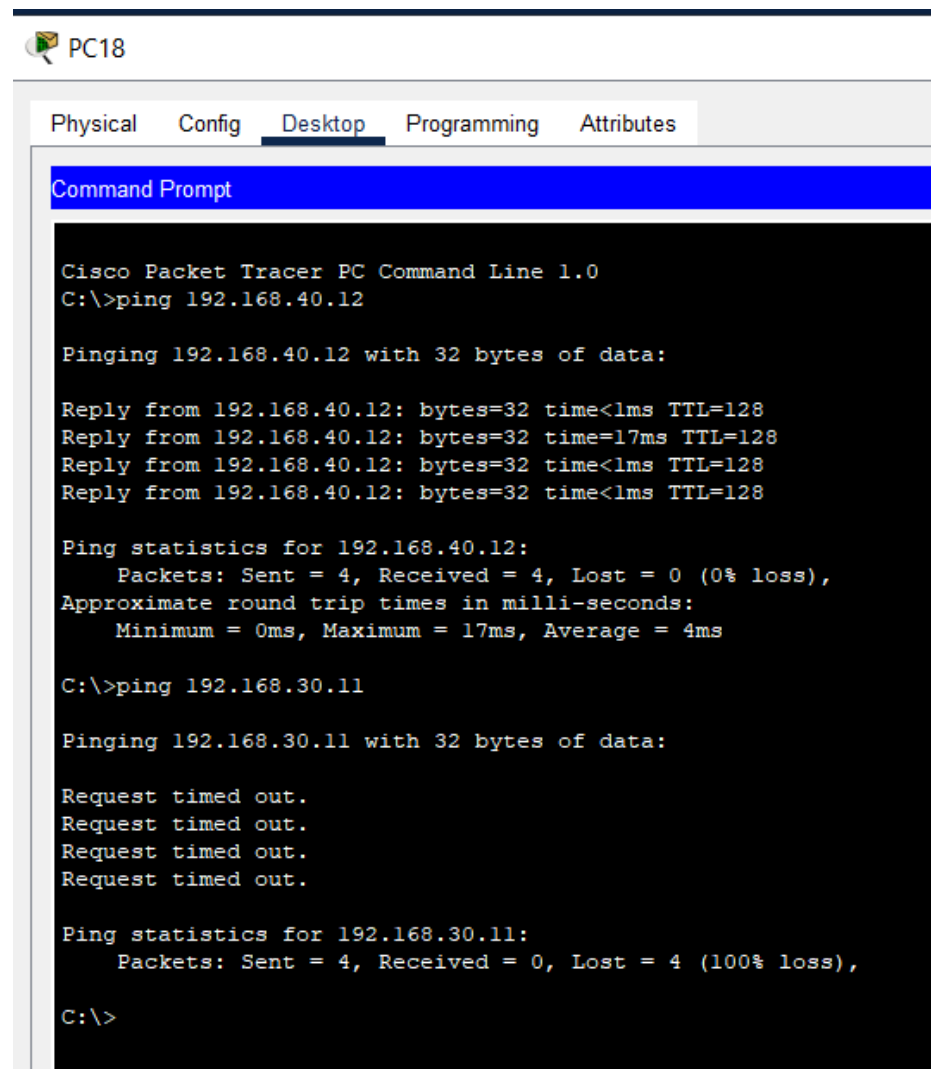
- PC18 → 192.168.40.10
 - PC19 → 192.168.40.11
 - PC20 → 192.168.40.12
 - PC21 → 192.168.40.13
- Mask: 255.255.255.0.

1ST PING:

WITHIN VLAN 40 SUCCESSFUL

2ND PING:

DIFFERNERT VLAN 40 TO VLAN 30 UNSUCCESSFUL



The screenshot shows the PC18 configuration window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.12

Pinging 192.168.40.12 with 32 bytes of data:

Reply from 192.168.40.12: bytes=32 time<1ms TTL=128
Reply from 192.168.40.12: bytes=32 time=17ms TTL=128
Reply from 192.168.40.12: bytes=32 time<1ms TTL=128
Reply from 192.168.40.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.40.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms

C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

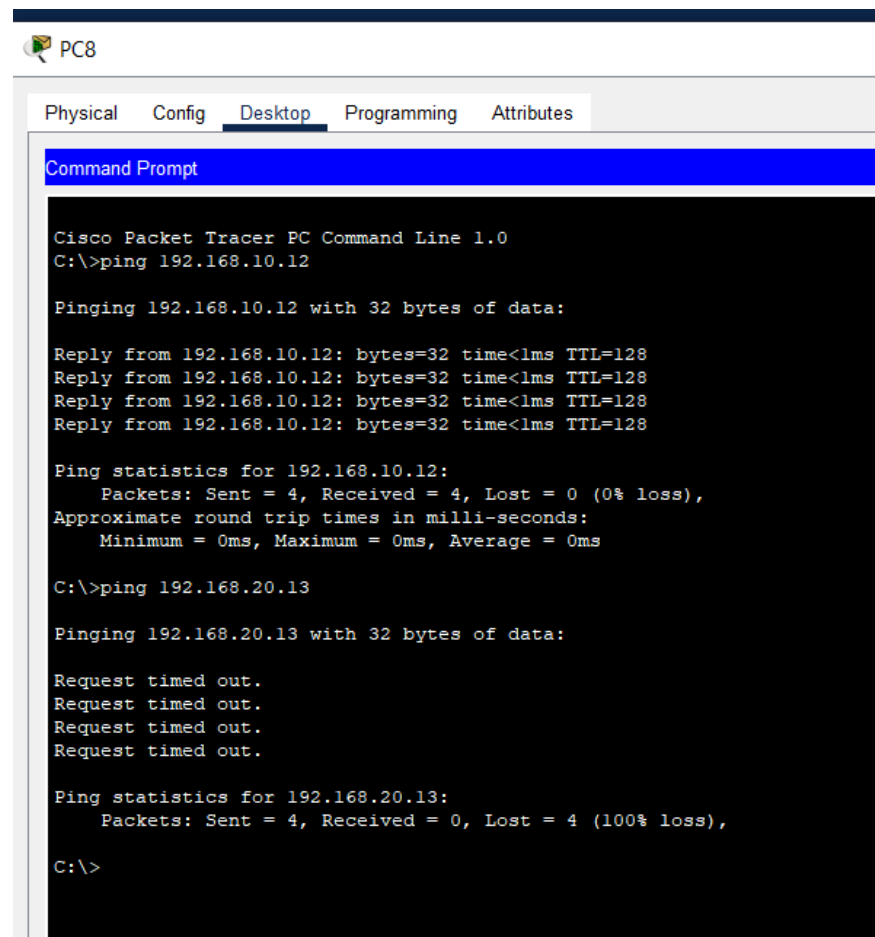
C:\>
```

1st PING:

SUCCESSFUL WITHIN SAME VLAN

2ND PING:

DIFFERENT VLAN UNSUCCESSFUL



The screenshot shows the 'PC8' configuration window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of two ping commands. The first command, 'ping 192.168.10.12', is successful, showing four replies with 0% loss. The second command, 'ping 192.168.20.13', fails, showing four 'Request timed out' messages and 100% loss.

```
PC8
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.13

Pinging 192.168.20.13 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.13:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```